A5: Angstrom Derivation and Model-Validity Plan

Assessment code: A5 (individual)
Student: Kapil Naidu (lab partner Samuel Cohen; the measurements are our team's)
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Repository: https://github.com/kapilnaidu1/phys39-naidu-cohen, branch main
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Markdown source: docs/assessments/a5_angstrom_derivation_plan.md

1. The rod equation and what κ and ν mean

Take a slice of the rod between x and x + dx, with cross-section $A = \pi R^2$, perimeter $P = 2\pi R$, density ρ, specific heat c, conductivity k and side heat-transfer coefficient h to a room at T∞. Heat stored = heat conducted in minus heat conducted out minus heat lost through the side:

$$\rho\, c\, A\, dx\, \partial T/\partial t = kA\, \partial T/\partial x\,|_{(x+dx)} - kA\, \partial T/\partial x\,|_{(x)} - h\, P\, dx\, (T - T_\infty)$$

Divide by ρcA dx, let dx → 0 and write θ = T − T∞:

$$\partial\theta/\partial t = \kappa\, \partial^2\theta/\partial x^2 - \nu\, \theta, \qquad \kappa = k/(\rho c), \qquad \nu = hP/(\rho cA) = 2h/(\rho cR)$$

κ is the thermal diffusivity (how fast a temperature change spreads along the rod) and ν is the side-loss rate (how fast a slice relaxes toward room temperature through its surface). The assumptions are: one temperature per cross-section (small transverse Biot number, section 6.1), constant k, ρ, c and h, linear side loss, constant room temperature, and the measured temperature at the first rod hole (T1, x = 0) as the boundary condition. At steady state the same equation gives $\kappa\theta'' = \nu\theta$, so the Module 7 decay constant is $m^2 = \nu/\kappa = hP/(kA)$.

2. Periodic solution

Drive the base as $\theta(0, t) = \theta_b \cos \omega t$ with $\omega = 2\pi/\tau$, and first treat the rod as semi-infinite ($x \geq 0$, $\theta \to 0$ far away). Guess a complex exponential, because differentiating it returns the same function times a constant:

$$\tilde{\theta}(x, t) = \theta_b\, e^{(-s\, x)}\, e^{(i\, \omega\, t)} \qquad \partial\tilde{\theta}/\partial t = i\omega\, \tilde{\theta}, \qquad \partial^2\tilde{\theta}/\partial x^2 = s^2\, \tilde{\theta}$$

Substituting into the rod equation and cancelling θ~ turns the PDE into algebra:

$$i\, \omega = \kappa\, s^2 - \nu \qquad \rightarrow \qquad s^2 = (\nu + i\, \omega)\, /\, \kappa$$

Keep the root with positive real part (the other one grows without limit) and write $s = q + iq'$. Then $s^2 = q^2 - q'^2 + 2iqq'$, and matching real and imaginary parts:

$$q^2 - q'^2 = \nu/\kappa \qquad 2\, q\, q' = \omega/\kappa$$
$$(q^2 + q'^2)^2 = (\nu^2 + \omega^2)/\kappa^2$$
$$q = \sqrt{[\ (\ \nu + \sqrt{(\nu^2 + \omega^2)}\ )\ /\ (2\kappa)\ ]} \qquad q' = \sqrt{[\ (\ -\nu + \sqrt{(\nu^2 + \omega^2)}\ )\ /\ (2\kappa)\ ]}$$

The physical temperature is the real part:

$$\theta(x, t) = \theta_b\, e^{(-q\, x)} \cos(\omega\, t - q'\, x)$$

Every point oscillates at the drive frequency. The amplitude falls by a factor e every 1/q, and the phase lags by q′x radians (a time delay q′x/ω). Two limits check the algebra: with no side loss (ν = 0) $q = q' = \sqrt{(\omega/2\kappa)}$; with side loss q > q′, because loss removes amplitude without adding delay. A shorter period (larger ω) raises both q and q′.

The setpoint mean is above room temperature, so the full solution also contains the stationary profile of Module 7. The equation is linear, so the two add; in the fits the mean is absorbed by a constant per sensor.

For the real rod, with an insulated top a distance L above T1, the periodic solution is

$$\tilde{\theta}(x, t) = \theta_b\, e^{(i\omega t)}\, \cosh(s(L - x))\, /\, \cosh(sL) = [\text{semi-infinite}] \times (1 + e^{(-2s(L-x))})\, /\, (1 + e^{(-2sL)})$$

The extra factor is a wave reflected from the top, of relative size about $e^{(-2q(L-x))}$. It is the finite-length check of section 6.2.

3. Inverse relations and dimensional checks

For two sensors $\Delta x = x_2 - x_1$ apart, $q = \ln(B_1/B_2)/\Delta x$ and $q' = \Delta\varphi/\Delta x$. With five sensors, q and q′ are the slopes of ln B and of the unwrapped phase against x. Rearranging the two matching conditions:

$$\kappa = \omega\, /\, (2\, q\, q') \qquad \nu = \kappa\, (q^2 - q'^2) \qquad k = \rho\, c\, \kappa = \rho\, c\, \omega\, /\, (2\, q\, q')$$

The conductivity needs only the two measured slopes, the drive frequency and ρc; h never has to be measured. That is the point of the method.

Quantity	Units	Check
θ, t, x	K, s, m	
$\partial\theta/\partial t$, $\kappa\partial^2\theta/\partial x^2$, $\nu\theta$	K/s each	$(m^2/s)(K/m^2) = K/s$; $(1/s)K = K/s$
s^2, q^2, q'^2	$1/m^2$	$(\nu + i\omega)/\kappa = (1/s)/(m^2/s)$
$\kappa = \omega/(2qq')$	m^2/s	$(rad/s)/(1/m^2)$
$\nu = \kappa(q^2 - q'^2)$	1/s	$(m^2/s)(1/m^2)$
$k = \rho c \kappa$	W/(m K)	$(kg/m^3)(J/(kg\ K))(m^2/s) = J/(s\ m\ K)$
$\nu = 2h/(\rho c R)$	1/s	$(W/(m^2\ K))\ /\ ((J/(m^3\ K))\ m)$

Numerical check with the course example (ω = 0.0628 rad/s, q = 25 m^{-1}, q′ = 18 m^{-1}): κ = 0.0628/(2 × 25 × 18) = 6.98 × 10^{-5} m^2/s and ν = 6.98 × 10^{-5} × (625 − 324) = 2.10 × 10^{-2} s^{-1}, as on the course page.

4. Sign and phase conventions

- x is measured from T1's hole (the first rod hole), positive up the rod toward the top. T1 is the boundary; the TEC thermistor and the setpoint are not.
- θ = T − T∞ with T∞ = 22.53 °C, the mean of T1 to T5 (calibration offsets applied) with control off and the rod unheated, 09:47 on 2 November.
- Each sensor is fitted as $T_i(t) = a_i + B_i \cos(\omega t + \varphi_i) + c_i(t - \bar{t})$, with $B_i > 0$. The drift term c_i takes up the slow warming of the mean. With this convention a sensor that lags has a smaller φ_i.
- Phases are unwrapped starting from T1 and going up the rod, so $\ln B(x) = \ln B_0 - qx$ and $\varphi(x) = \varphi_0 - q'x$, both slopes negative and q, q′ > 0.
- The Arduino setpoint is $T_a + T_b \sin(\text{phase})$, which is a cosine shifted by −π/2. Only phase differences between sensors enter q′, so the offset cancels.
- Calibration offsets (26 October room cross-check) are subtracted in analysis. A constant offset changes only a_i, not B_i or φ_i.
- Errors: B_i and φ_i from the scatter of the same fit done cycle by cycle (standard error over 5 cycles), combined with a ±1 mm position uncertainty for T2 to T5 (T1 defines x = 0). Slope errors are scaled by $\sqrt{(\chi^2/\text{dof})}$ when that exceeds 1.

5. Acquisition plan

Drive. T_a = 40 °C, T_b = 16 °C (24 to 56 °C requested), PI on the TEC thermistor with K_p = 12, K_i = 0.2 and conditional-integration anti-windup, and negative commands multiplied by $r = \chi_h/\chi_c = 0.501/0.181 = 2.771$ (Module 4) before the ±255 clamp. In the S18 comparison at 100 s, open loop gave a TEC swing of 23 % of T_b, P 76 % with a 2.6 °C droop, and PI 87 % with no droop and no saturation, so PI is the production mode. T_a and T_b stay fixed for every period so that only the frequency changes.

Periods. Planning numbers below use κ and ν from the S18 finite-rod fit at 480 s (section 6.3) and the sensors at 0 to 101.6 mm:

Period (s)	1/q (mm)	predicted B5/B1	predicted lag T1 → T5 (rad)	finite-length change of B5/B1	records per cycle
100	54	0.152	1.83	+0.1 %	200
180	71	0.240	1.34	0.0 %	360
240	82	0.285	1.14	−0.8 %	480
300	90	0.319	1.01	−1.8 %	600
480	111	0.386	0.80	−3.4 %	960

The course asks for the farthest sensor near B5 ≈ B1/e = 0.37 at the primary period, a T1 to T5 lag of 0.5 to 2 rad, and no period below 100 s. **Primary period 480 s**: measured on S18, B5/B1 = 0.385 and 0.81 rad, both on target, and the TEC follows the setpoint at 99 % amplitude. **Second period 100 s**: the course minimum, with measured B5/B1 = 0.149 and 1.79 rad; T5 still swings 0.43 °C, about 27 times its 0.016 °C fit residual. **Third period 240 s** for S20 sits between them and tests whether κ depends on frequency.

Sampling. One record every 0.5 s, each the mean of 1000 ADC readings (TEC) or 500 readings (each rod channel). That is 200 to 960 records per cycle, far more than a sinusoid needs; the averaging keeps the fit residuals at 0.016 to 0.03 °C at 100 s.

Settling criterion (fixed before looking at the fits). Fit each complete cycle on its own. A period counts as settled from the first cycle after which every sensor's B stays within 3 % and φ within 0.05 rad of the cycle before, with no steady trend over three cycles. The mean may still creep (the drift term handles it). Keep at least 5 complete settled cycles. On S18, PI at 100 s met this from its second cycle; at 480 s the first cycle after the switch had a TEC swing 3.5 % low and was excluded.

Run lengths for S20. About 2 cycles to settle plus 5 kept: 480 s about 56 min, 240 s about 28 min, 100 s about 12 min, roughly 1 h 40 min in one scheduled run with phase-continuous transitions kept in the record.

Sensors. TEC thermistor in the base; T1 to T5 taped into the first five holes at 0, 25.4, 50.8, 76.2 and 101.6 mm (±1 mm, `data/module_08/sensor_map.csv`); rod diameter 12.757 ± 0.006 mm by calipers; L = 281.9 mm from T1's hole to the top (`data/module_08/rod_geometry.csv`). No sensor moves between periods.

Safety. Software latch at 56 °C on T1 (the course limit for the rod base) and 65 °C on the TEC thermistor; the 70 °C thermal switch stays as the hardware backstop; setpoints are refused outside 10 to 56 °C; no period below 100 s. Highest raw readings on S18: T1 41.3 °C, TEC 55.9 °C.

Room temperature. Measure before heating and again after the run with everything at rest. On S18 it was measured only before, which is the main limitation on the means a_i but does not affect B_i or φ_i.

6. Model-validity checks

6.1 Transverse Biot number

The 1-D equation needs the cross-section to be nearly isothermal, $\mathrm{Bi}_\perp = hR/(2k)$ much less than 1. Because $\nu/\kappa = hP/(kA) = 2h/(kR)$, the measured slopes give it without knowing h or k: $\mathrm{Bi}_\perp = (\nu/\kappa)R^2/4$. With R = 6.378 mm:

Source	ν/κ (m⁻²)	$\mathrm{Bi}_\perp$
Angstrom, 480 s finite-rod fit	21.7 ± 3.8	2.2×10^{-4}
Angstrom, 100 s	41.6 (ν poorly determined)	4.2×10^{-4}
Module 7 steady profile, m²	16.8	1.7×10^{-4}

All are below 10^{-3}, against my tolerance of 0.01, so the radial temperature variation is negligible. The 480 s value of ν/κ agrees with the independent Module 7 steady profile within 1.3 standard errors. With handbook ρc for aluminium (2.42×10^6 J/(m³ K)), $\nu = 2.04 \times 10^{-3}$ s⁻¹ corresponds to $h \approx \nu\rho cR/2 \approx 16$ W/(m² K), a normal value for still air.

6.2 Finite length

Using the measured $s = q + iq'$ at each period, the finite-rod factor of section 2 changes the ratios B_i/B_1 and the lags $\varphi_1 - \varphi_i$ by:

Period		T2	T3	T4	T5
100 s	B_i/B_1	0.0 %	0.0 %	+0.03 %	+0.13 %
100 s	phase (rad)	0.000	0.000	0.000	0.000
480 s	B_i/B_1	−0.4 %	−1.0 %	−2.0 %	−3.1 %
480 s	phase (rad)	0.002	0.001	−0.003	−0.016

With a 1 % tolerance, the semi-infinite form is fine for all five sensors at 100 s, but at 480 s only T1 to T3 meet it. At 480 s I therefore fit the insulated-tip finite-rod solution to all five sensors instead of dropping T4 and T5.

6.3 Data-based checks (S18)

- **Boundary waveform.** At T1 the second harmonic is 2.3 % (100 s) and 1.5 % (480 s) of the fundamental, the heating and cooling halves agree within 0.2 °C (100 s comparison), and the PI command never reached the clamp. T1 is a clean sinusoid.
- **Sign of ν.** $q > q'$ at both periods, so $\nu > 0$, as side loss requires.
- **Spatial residuals.** The straight-line fits have χ^2/dof of 3.3 and 4.7 (100 s) and 2.3 and 3.2 (480 s): the points scatter about twice their errors, so a sensor-to-sensor effect of a few percent (contact or position) is not yet in the error budget.
- **Frequency consistency**, `data/module_08/angstrom_finite_angstrom_live_20261102_094506.txt`:

Fit	κ at 100 s (10^{-5} m²/s)	κ at 480 s (10^{-5} m²/s)	difference	χ^2 at 480 s (6 dof)
semi-infinite	9.74 ± 0.23	8.93 ± 0.25	2.4 σ	16.8
finite rod, L = 281.9 mm	9.74 ± 0.23	9.40 ± 0.24	1.0 σ	11.1

The semi-infinite fits disagree between the two periods; the finite-rod fits agree within one standard error and fit the 480 s data better. The frequency dependence is the finite-length effect that section 6.2 predicts.

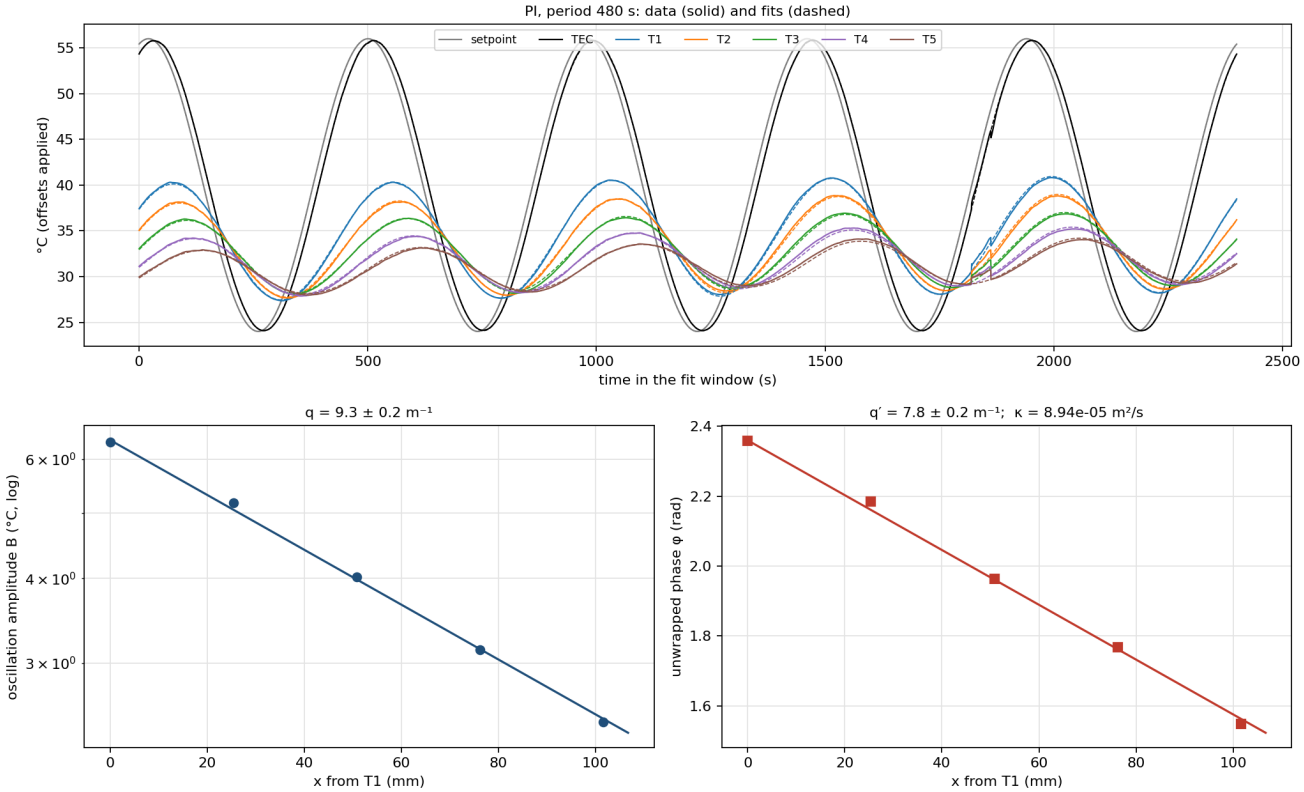


Figure 1. PI at 480 s, cycles 2 to 6 of the S18 run. Top: setpoint, TEC and T1 to T5 with the per-sensor fits (dashed). Bottom: amplitude on a log scale and unwrapped phase against position, with the semi-infinite lines (q = 9.34 ± 0.18 m^{-1}, q′ = 7.84 ± 0.15 m^{-1}).

- **Comparison with the literature.** The handbook values for 6061-T6 (k = 167 W/(m K), ρ = 2700 kg/m^3, c = 896 J/(kg K)) give κ = 6.9 × 10^{-5} m^2/s, about 30 % below our finite-rod values. Neither the Biot number nor the finite length explains this. Things to test in S19 to S22: the sensor positions (κ scales as Δx^2, so a 1 mm error in each 25.4 mm spacing moves κ by 8 %; the bead may not sit at the hole's centre line), different response lags of the taped sensors, and the alloy itself.

7. Conclusion: do the retained sensors support the 1-D analysis?

Yes, with one condition. Across the rod, $Bi_\perp \le 4 \times 10^{-4}$, so every cross-section is at one temperature and T1 to T5 measure the 1-D field. Along the rod, the semi-infinite solution is adequate for all five sensors at 100 s (finite-length change ≤ 0.13 %), but at 480 s it is off by 2 to 3 % at T4 and T5; there I use the finite-rod solution with L = 281.9 mm, which brings κ at the two periods into agreement (9.74 ± 0.23 and 9.40 ± 0.24 × 10^{-5} m^2/s). The remaining gap to the handbook κ and the spatial χ^2 of 2 to 5 are the open questions for the production run, not signs that the one-dimensional picture fails.

8. Files and commands

- Live run: `data/module_08/angstrom_live_20261102_094506.csv` (with `_events.txt`, `_notes.csv`, `_meta.json`); sketch `arduino/module_08/angstrom_tec/angstrom_tec.ino`; GUI `python/angstrom_gui.py`.
- Notes: `docs/module_notes/module_08_angstrom.md`.

```
RUN=data/module_08/angstrom_live_20261102_094506.csv
CAL=data/module_08/calibration_rod_run_20261026_112627.csv
python3 python/analysis/angstrom_fit.py $RUN --mode PI --period 100 --cycles 5 --offsets $CAL
python3 python/analysis/angstrom_fit.py $RUN --mode PI --period 480 --cycles 5 \
    --t-end 5507.21 --offsets $CAL
python3 python/analysis/angstrom_fit.py $RUN --compare --period 100 --cycles 3 --offsets $CAL
python3 python/analysis/angstrom_finite_fit.py \
    data/module_08/angstrom_fit_angstrom_live_20261102_094506_PI_100s.csv \
```

```
data/module_08/angstrom_fit_angstrom_live_20261102_094506_PI_480s.csv \
--length-mm 281.9 --dx-mm 1
```

9. AI use

I used an AI assistant (Claude) to help draft this write-up, the analysis scripts and the figures from our team's measurements, as the course AI policy allows. I checked the derivation step by step and checked every number against the scripts and data files listed above.